

Chemical Strengthening of Glass: Lessons Learned and Yet To Be Learned

Arun K. Varshneya*

Saxon Glass Technologies Inc., Alfred, New York

NY State College of Ceramics at Alfred University, Alfred, New York 14802

This paper reviews the progress that has been made in our understanding of the chemical strengthening of glass by ion exchange over its nearly five decades of history. Lessons learned are briefly discussed; more importantly, those which are yet to be learned are highlighted. It is recognized that, except for detailed compositional effects, the kinetics of ion interdiffusion process and the chemical strengthening technology are reasonably well understood. However, the science of stress generation and its concurrent relaxation is far from being clear despite the elegant analogy to thermal stresses invoked by Cooper. The need to understand plasticity of glass network during accommodation of a larger invading ion is emphasized. In turn, the influence of network topology on its yield strength in shear as well as hydrostatic modes is recognized. For expanded applications under extreme conditions of loading, damage evolution in chemically strengthened glass needs to be studied. Such a study is linked to our understanding of the terms “strength,” “hardness,” “toughness,” and “brittleness” of glass.

Introduction

By some estimates (see Varshneya,¹ p. 479), the theoretical strength of a flawless glass is as much as 35 GPa applied in a tensile mode. On the other hand, ordinary soda-lime glass products break in a brittle mode at as little as 7 MPa (~ 1000 psi) tension. Containers and windows readily fracture at 35–50 MPa. Optical fibers, because of their relatively small diameter, are an example of a rare unstressed product that

withstand ~ 700 MPa or more. Despite utmost care, monolithic pieces of glass do not withstand much of applied tension, primarily because of its interaction with the surface flaws of the order of a few micrometers (often termed as “Griffith flaws”), produced during handling by the manufacturer as well as the user in ordinary ambient conditions. Flaws propagate under the action of the stress intensity factor K_I at the atomically dimensioned (sharp radius of curvature) tip of a flaw and are assisted by relative humidity through a stress-assisted corrosion mechanism. When K_I approaches K_{IC} , the critical stress intensity factor, a spontaneous fracture is imminent. Almost any material contact abrades a pristine surface of glass to produce what could be a fatal flaw. In contrast, ductile metals are easily able to with-

*varshneya@alfred.edu

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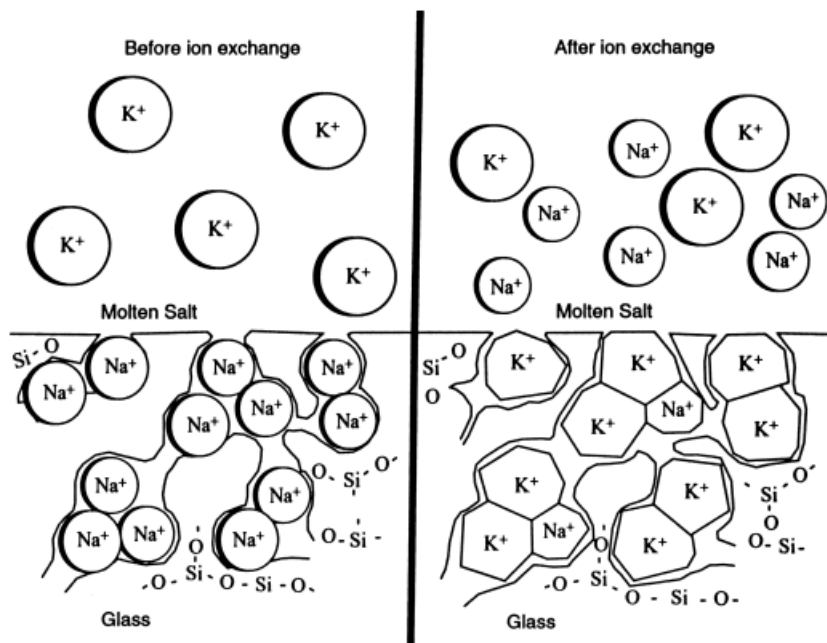


Fig. 1. Schematics of ion exchange.

stand as much as 150–350 MPa, and plastic products do not break because of their high-energy absorption capability (toughness). Indeed, had it not been for the poor usable strength characteristics, glass would have been the material of choice by virtue of its transparency, its inertness to most fluid media, and its perceived esthetics.

A variety of glass strengthening techniques can be used. These are briefly discussed by Varshneya¹ (p. 513ff). The underlying concepts are basically five:

- (1) Reduce the severity of a surface flaw (reduce K_I).
- (2) Control of the environment in the immediate vicinity of the crack tip (increase K_{IC}).
- (3) Polymeric coatings to reduce surface abrasion (reduce the size of the flaws).
- (4) Introduce surface compression (applied tension needs to overcome the existing surface compression).
- (5) Crack pinning and crack deflection around a second-phase precipitate (increases toughness, and hence reduces the energy available for the propagation of the crack).

Fire polishing and etching to accomplish (1) above are merely temporary means, mostly utilized by manufacturers to reduce fracture probability during processing. Introduction of surface compression remains generally the easier and the more cost-effective means of glass strengthening.

Common methods of introducing surface compression are thermal tempering, chemical strengthening (also called “ion exchange strengthening”), encasing with a lower thermal contraction coefficient material (usually a glaze), and surface crystallization. Of these, the commercial strengthened glass market is essentially dominated by thermal tempering because of its relative ease of application (hence, its cost-effectiveness) and its relative forgiveness to handling flaws due to mostly, for instance, the nearly 1 mm “case depth” (depth of compression layer) in a 5 mm wall thickness. Chemical strengthening is a distant second.

In chemical strengthening, an alkali-containing glass is immersed in an electrically heated bath (tank) containing a molten alkali salt such as KNO_3 at temperatures well below the glass annealing point. An ion exchange between the host alkali ions of glass and invading ions from the salt occurs (see Fig. 1). If the invading ions are larger in size than the host ions, then the resultant stuffing in a near-rigid atomic network of glass leads to the development of high surface compression and some balancing interior tension depending on the diffusion depth and product wall thickness.

The advantages of this process are: (i) introduction of relatively high surface compression, (ii) no measurable optical distortion, (iii) thin plates, even 100 μm

thin, can be strengthened, and (iv) irregular geometry products can be readily strengthened so long as the surface can be contacted by the molten salt.

Disadvantages of chemical strengthening are (a) specifically limited to alkali-containing glass, (b) low case depth for soda-lime glass that makes the glass susceptible to weakness from common handling flaws, and (c) high cost due to extended bath immersion.

The reader is advised to refer to several reviews^{2–6} for a comprehensive understanding of the glass chemical strengthening process.

In soda-lime glass products, immersion in KNO_3 salt bath at $\sim 475^\circ\text{C}$ for 16 h can develop as much as ~ 450 MPa surface compression, but a case depth of only about 25–30 μm . The interior tension can be < 4 MPa for a 5 mm plate, which means that the glass can be cut afterwards (the exposed edges, however, are no longer strengthened and, hence, must be protected).

Chemical strengthening technology works best for lithium and/or sodium aluminosilicate glasses, is of marginal use for Corning Code 7740 (Pyrex[®]) type borosilicate, and “lead crystal” glass compositions, and is of no use for fused silica glass. However, “Type I neutral glass” such as Kimble N51A medium expansion borosilicates used in pharmaceutical syringes, ampoules, and vials can be strengthened reasonably well with care. The best one can achieve currently in a single-step strengthening process is demonstrated by the author⁷ for a high T_g lithium aluminosilicate glasses immersed in mixed $\text{NaNO}_3/\text{KNO}_3$ baths. Surface compression as high as ~ 1 GPa (145,000 psi) decreasing rapidly to ~ 280 MPa at ~ 40 μm , then slowly decreasing to zero at ~ 1 mm depth has been developed.

Chemical strengthening was discovered by Kistler⁸ and Acloque and Tochon⁹ in 1962. The initial excitement decayed rapidly after it was recognized that ordinary soda-lime glass products are slow to develop a 25 μm case depth even after 4–24 h of immersion. In more recent years, the emphasis on product safety and environmental concerns on the use of fossil fuel products have renewed interest in the process. In some cases such as autoinjectors made using Type I borosilicate glass, chemical strengthening developed by Saxon Glass Technologies Inc. is the enabling technology that has made the difference between market acceptability and otherwise a risky product. Examples of other commercially successful glass products that use chemical strengthening are aircraft cockpit windshields, photocopier scanner glass, display windows in

mobile personal electronic devices (e.g., cell phone and MP3 players), and compact disks for portable hard drives.

A large variety of glass products could, however, be strengthened and brought to market at an acceptable premium. Use of glass for transparent armor, hurricane-resistant architectural windows, solar energy conversion tubular and flat-plate collectors, panels for large displays, and needle-free drug delivery cartridges are perhaps around the corner. Pressurized beverage containers could be lightweighted and strengthened to make recycling a cost-effective option. Relative to steel, highly strengthened glass has a higher strength-to-weight ratio, and hence could also be used in support structures for architecture and in wide-spread use of thinner, yet stronger, glass windows.

The objective of the present article is to take an inventory. Where are we in terms of the science and technology: what do we know about the chemical strengthening process, and what do we need to know in order to expand its use for a wider applicability? The reader is strongly advised to review the concepts of glass strength outlined in the inaugural issue of this journal.¹⁰

The Science of Chemical Strengthening

Chemical strengthening of glass is comprised of three components:

- (1) Kinetics of alkali ion exchange: To generate a deep case depth, the ion exchange process should be as rapid as possible.
- (2) Generation of stress and its concurrent relaxation: compression on the outside, tension on the inside. For a successful application, surface compression generally needs to be maximized and its relaxation should be as little as possible.
- (3) Damage evolution: How does surface compression translate into meaningful strength increase for an intended application? This topic includes an understanding of hardening, toughening, and brittleness and their relationship to strengthening.

Kinetics of Alkali Ion Exchange

The topic of ion exchange kinetics is actually well understood. The concentration profile of the invading ion, hence that of the host ion (from an ion-for-ion exchange calculation) is described by an interdiffusion coefficient $\bar{D}$, and given by the Nernst–Planck expression

(ignoring second-order effects of thermodynamic activity):

$$\bar{D} = \frac{D_{\text{Na}}D_{\text{K}}}{D_{\text{Na}}N_{\text{Na}} + D_{\text{K}}N_{\text{K}}} \quad (1)$$

where N_i is the fractional concentration of alkali ion i and D_i is its self-diffusion coefficient in mixed-alkali glass compositions, and not in the parent glass. As the self-diffusion coefficients change over several magnitudes in mixed-alkali glass compositions (another manifestation of the familiar “mixed-alkali effect”), the interdiffusion coefficient is concentration dependent and is essentially controlled by the slower moving specie (see Fig. 2 taken from Varshneya and Milberg¹¹). This unfortunately slows down the strengthening process (when conducted without electric or sonic field assistance). An additional impeding mechanism is the effect of the generated stresses themselves on the kinetics. It is now well recognized^{12,13} (Tomozawa, unpublished data) that the $\bar{D}$ is impeded by the surface compressive field and accelerated by the interior tensile field. One can understand that the stresses expand or constrict an average interstitial ion site.¹⁴ Because the glass surface determines the rate of intake of the invading ions, the overall diffusion rate is indeed helplessly slow.

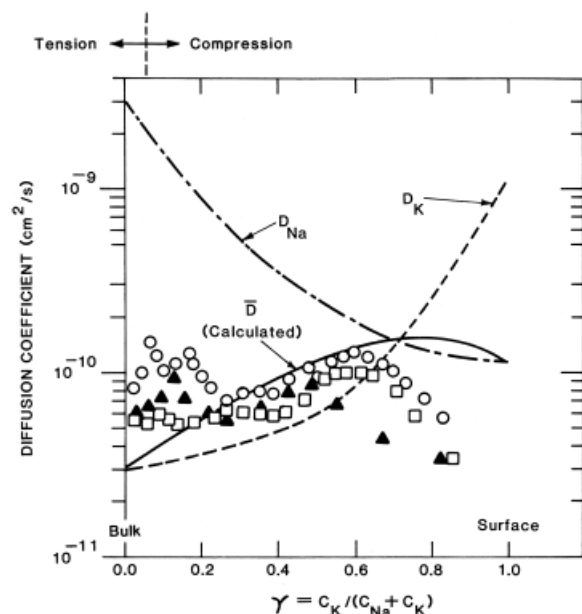


Fig. 2. Variation of the interdiffusion coefficient $\bar{D}$ vs c for a borosilicate glass. After Varshneya and Milberg.¹¹

How can the rate of exchange be increased?

There are several means of achieving this:

(1) Use mixed-alkali compositions: Figure 2 shows that $\bar{D}$ is larger for mixed-alkali compositions.

(2) Use lesser size-disparity alkali pair for exchange, for example, Na^+/Li^+ exchange is much faster relative to K^+/Na^+ exchange.

(3) Use alkali aluminosilicate glasses with an alkali/alumina ratio of ~ 1 , where the high parent glass transition temperature and reduced nonbridging oxygens allow faster and deeper exchanges at higher immersion temperatures.^{7,15–18}

(4) Use mixed-salt immersion baths.^{7,19}

(5) Use electric-field assistance²⁰: This method is essentially based on electrolysis, except that the large alkali ions are forced electrically to enter the glass network from side “1,” which made the anode plate, displacing the host alkali ions from side “2,” which made the cathode plate. A reversed current flow may be needed to allow the invading ions enter from side “2” for about half the time to obtain a nearly symmetric concentration profile of the invading ions.²⁰

(6) Use sonic assistance²¹: A sonotrode attached to the specimen can accelerate the ion exchange. Apparently, this occurs by periodic dilatation of the interstitial sites for a long enough time that allows an ion to get in farther.

(7) Add minor quantities of MgO in the glass composition.²²

For (1) and (2), whereas the case depth could be improved this way, the magnitude of surface compression may be penalized in either of these approaches. Technique (3) is well recognized and is limited to aluminosilicate glasses, and hence of limited commercial interest. The fourth technique requires an $\text{Li}/\text{Na}_2\text{O}$ -containing parent glass immersed in mixed $\text{NaNO}_3/\text{KNO}_3$ salt bath. Essentially, K^+ from the bath replaces Na^+ in the glass over a short distance and the Na^+ from the bath replaces Li^+ in the glass over a larger penetration distance. The primary issue in the field-assisted exchange experiment is the electric isolation of the two sides. If this were to be achieved only by keeping a portion of the specimen out of the salt contact, then the unexchanged region acts as an Achilles’ heel. A high-magnitude tension spike is likely to be generated by stress reversal just above the exchange line, which could result in glass fracture. The sonic assistance technique has limited appeal; what accelerates the diffusion also accelerates the relaxation of the generated stress. Thus,

case depth is higher at the expense of the magnitude of the surface compression. Finally, the last technique, *adding small amounts of MgO to the glass composition is perhaps the more promising of the methods listed here.* The method, based on an extensive study by LaCourse's students,²² requires that if container and flat glass products are to be chemically strengthened better than what is currently achievable, then we need to know the spectrum of minor additions to the typical soda-lime composition that would augment the interdiffusion coefficients.

Stress Production and Relaxation

The science of stress generation in glass during ion exchange is described by Cooper and Krohn²³ by analogy to thermal stresses. For a semi-infinite (in y - and z -dimensions) glass plate of a thickness of $2L$ being penetrated from the $\pm x$ -direction, the stresses at a location x are given by

$$(\sigma_{xx})_x = 0$$

$$(\sigma_{zz})_x = (\sigma_{yy})_x = -\frac{BEC_x}{(1-\nu)} + \frac{1}{L} \int_0^L \frac{BEC_x}{(1-\nu)} dx \quad (2)$$

where E is the Young's modulus; ν the Poisson's ratio; and $B = (1/3V) (\partial V/\partial C)$ the linear network dilatation coefficient (called the "Cooper coefficient" in honor of the late Alfred R. Cooper) [correction added after online publication May 28, 2010: corrected B definition]. Like the linear thermal expansion coefficient, the linear network dilatation coefficient is $1/3$ of the relative change in the volume V per unit exchange of concentration C . Essentially, the stress is obtained by replacing temperature T by concentration C , and the linear thermal expansion coefficient α by the linear network dilatation coefficient B in all of the thermal stress equations. An example of a concentration profile and a corresponding (measured) stress profile in a sodium borosilicate glass is shown in Fig. 3, from Varshneya.²⁴ The glass surface has the highest compression. Thereafter, compression decreases rapidly to zero and converts to a mild tension in the interior. The distance from the surface to the zero stress is the case depth.

Although there appears to be a general agreement between the stress and concentration shown in Fig. 3, there are three notable discrepancies:

(i) An anomalous tension maximum not predicted by Eq. (2) above usually exists. Schaeffer and Heinze²⁵

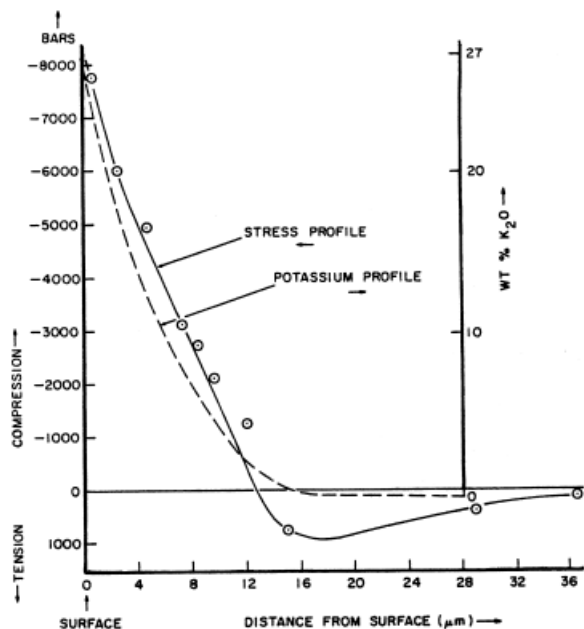


Fig. 3. Invading ion concentration and stress versus depth below surface for a borosilicate glass. After Varshneya.²⁴

suggested that the tension maximum was a result of measuring birefringence, which relates to $(\sigma_{yy} - \sigma_{xx})$ and not σ_{yy} ; the σ_{xx} (as per Eq. (2)) is zero only at the extreme outside surface, but, is nonzero subsurface in most finitely sized specimens. Using finite-element simulation, Varshneya and Petti²⁶ showed that, although there was some support to the above argument, the entire magnitude of tension maximum could not be explained. In yet another elegant and thought-provoking argument, Sane and Cooper²⁷ suggested that the anomalous tension maximum below the compression was an experimental artifact of slicing. Again, not all of the discrepancy could be explained in this manner.

(ii) A pronounced subsurface compression maximum, also not predicted by Eq. (2), is present in most ion-exchanged soda-lime glasses (see Fig. 4). Early suggestions for its origin have been viscous flow. It is logical to assume that stress generation near the glass transition range must compete (successfully for strengthening) against relaxation of the same stress by viscous flow (assuming that the mixed-alkali layers in the surface had significantly lower glass transition temperatures). However, Sane and Cooper²⁸ were *not* able to account for all of the surface stress distribution by incorporating reasonable rates of viscous flow. It is likely that such

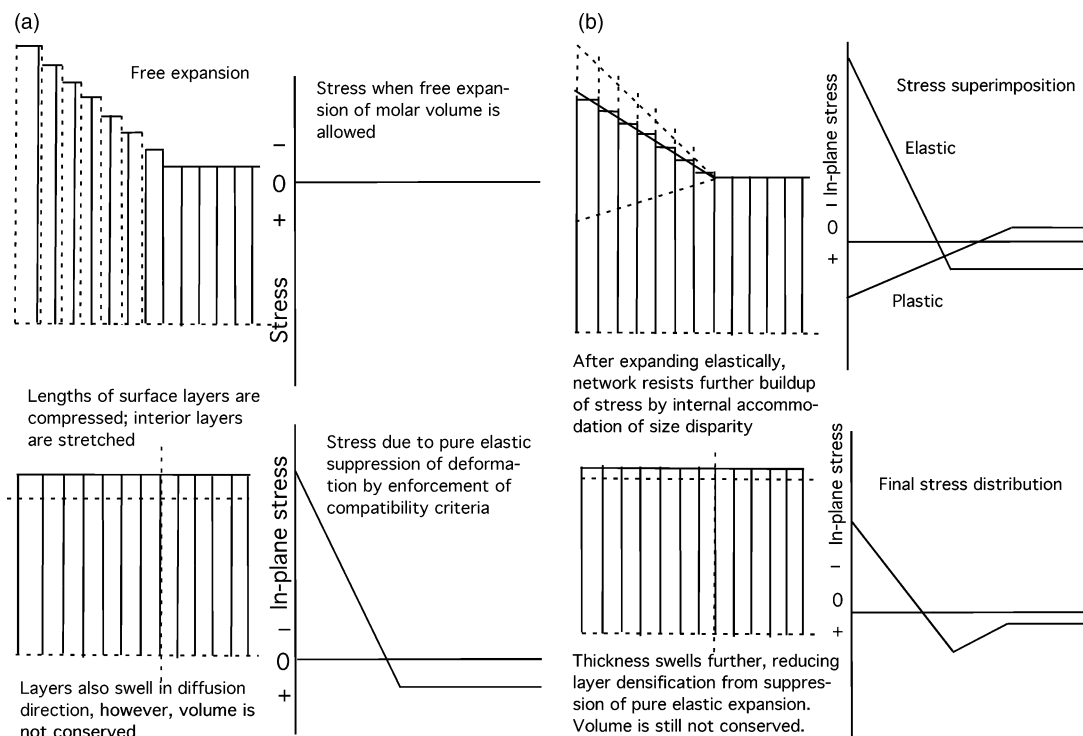


Fig. 5. Effect of network plasticity on stress buildup during ion exchange. If the body remains fully elastic for all of the incompatible strains produced by ion size disparity, then the suppression of these strains due to enforcement of elastic compatibility would give rise to high surface compression and interior tension (Fig. 5a). However, a portion of the size disparity is accommodated by shear and hydrostatic yielding. Hence, the overall incompatible strain is lesser; the net effect is the superimposition of the stress due to elasticity and a reverse stress due to plasticity (Fig. 5b).

even at the exchange temperature, and hence could not be ascribed the relatively high thermal contraction coefficient of the surface layers. The origin of this large de-

crease in stress magnitude and eventual changeover is not understood.

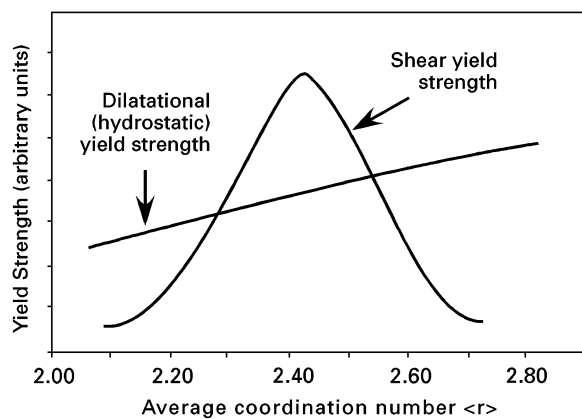


Fig. 6. Hypothetical variation of hydrostatic yield strength σ_{yh} and shear yield strength σ_{ys} with the average coordination number $\langle r \rangle$ for chalcogenide glasses. After Varshneya.³⁴

Damage Evolution

The increased demand for bullet- and blast-resistant and hurricane-resistant glass transparencies in the recent years requires a better understanding of how damage in glass occurs under extreme conditions of loading. The term "strengthening" implies an increase in the practically observable strength, that is, reduced susceptibility to fracture. After nucleation by some means, a flaw grows under the action of the tensile stress at the flaw tip to form a visible crack. At what point do we consider the resulting fracture to constitute sufficient damage in an intended application? The susceptibility of glass to damage must be different for the case of static loading versus the case of an indenter, where the tensile stress can always exceed any intrinsic strength at a sharp point of contact versus the case of a

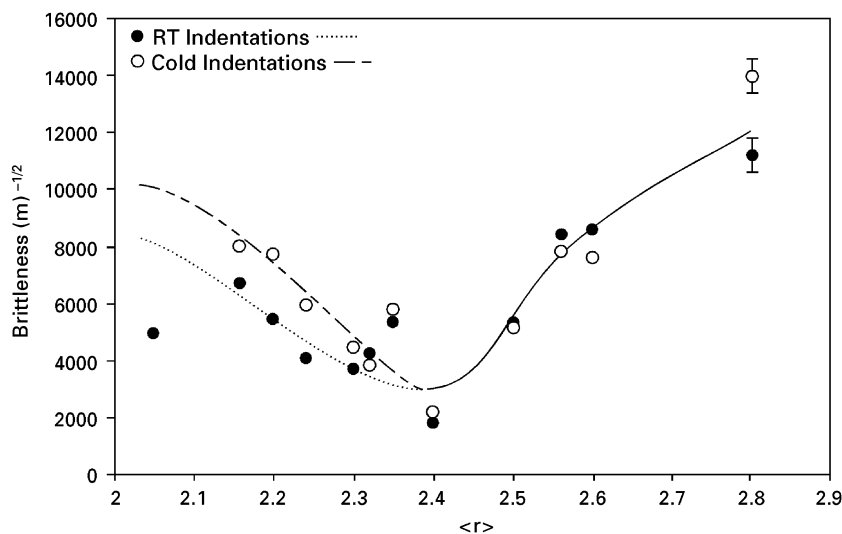


Fig. 7. Measured brittleness of Ge–Sb–Se chalcogenide glasses as a function of the average coordination number $\langle r \rangle$. After Varshneya *et al.*³⁸

high-velocity projectile causing high strain rates. Studies of damage evolution in glass are being left for a later publication. The following section on the underlying science of the various related terms in use helps to get the discussion started.

Stress, Strengthening, Hardening, Toughness, and Brittleness

Preexisting surface compression by ion exchange clearly act to impede the subcritical growth of a flaw, and hence “strengthening” of an article results. Because the ion exchange process introduces compression in the surface, which acts as a closure stress to flaws, new flaws could be more difficult to create. In other words, the abrasion resistance of glass may also increase somewhat. Likewise, the area under the stress–strain curve would increase making the glass somewhat tougher. However, increases in hardness and toughness are by and large much smaller compared with the increase in the strength, hence, the use of the names “chemical hardening” or “chemical toughening,” in place of “chemical strengthening” should be avoided.

The term “brittleness” (i.e., hardness/toughness) is often used to denote an increased susceptibility to brittle fracture. How does chemical strengthening affect brittleness? One may draw an analogy of the ion exchange process to indenting, where the invading indenter dis-

places the network elastically at first, and subsequently deforms the network plastically after exceeding the network yield strengths. Hence, the above discussion in “stress production and relaxation” relating deformations to glass network topology applies. Topologically optimized networks are expected to result in a greater hardness number (elastic recovery would be greater, hence, the residual indent is smaller). Elsewhere, Varshneya *et al.*³⁸ have shown that, in chalcogenide glasses, the optimally connected glass network (average coordination number = 2.4) also has the lowest brittleness (see Fig. 7). We may, therefore, be looking at a “triple benefit” of an optimally connected glass network: first, the brittleness is the lowest; second, the scratch resistance is increased; and third, the beneficial compression generated by ion exchange could be the highest in magnitude. *Thus, it is possible that topologically optimized alkali silicate glass network that has been chemically strengthened possesses the lowest tendency to fracture and remains the more robust of a product with respect to handling by a consumer.*

Measurement of Stress and Strength

Stresses are most conveniently measured using optical birefringence techniques. For flat glasses, a slice across the thickness with proper edge protection needs

to be cut, ground, and polished using standard ceramographic techniques. The thickness of the slice as cut could be 2–4 mm; however, because of high-stress magnitudes, the thickness may have to be reduced with care to 300 μm or less. Specimens can be examined for the optical birefringence using a polarizing light microscope or while immersed in nearly matching refractive index oil (mineral oil obtained from the local pharmacy usually works). Quantitative measurements of birefringence as a function of depth below the surface can be conveniently made using standard birefringence compensators such as a Babinet–Soleil wedge, a Berek compensator, or a Sénarmont compensator (as an accessory for the microscope). The measured birefringence is converted to stress using the familiar stress–optic law (see Varshneya,¹ p. 550ff). One should bear in mind that the birefringence corresponds only to the difference of principal stresses as stated above.

Surface compression and case depth can also be measured using ellipsometry in those specimens that have a suitable refractive index gradient within the diffusion zone. The “tinned” surface of float glass specimens happens to be ideal for such a measurement. Internal tension can be measured using laser interferometry.

For manufacturing quality control, strength can be measured using a variety of techniques such as four-point bending, biaxial flexure (ring-on-ring and ball-on-ring), and crushing by compression between parallel plates. These methods measure the “modulus of rupture” or the “MOR.” The ring-on-ring (ASTM C1499-04) or the ball-on-ring (ASTM F394-78) strength testing is particularly recommended as it is edge condition independent. Occasionally, QC may also be conducted using specialized methods such as steel ball drop (for displays in electronic devices), hydrostatic burst pressure, bird-shot impact test (for aircraft windshields), and “edge-on-impactor” (for transparent armor) tests. Also, any of the above methods could be used along with a standardized abrasion (AMOR). One must bear in mind that the usual equations that relate the applied load to strength are applicable only to the “small-deflection” scenario. Linear elasticity breaks down, and the expressions must be corrected for load-deflection non-linearity in most chemically strengthened glasses due to the large deflections from high-magnitude applied loads before fracture.

Because of the variability of strengths originating from specimen preparation techniques (the flaw condition of the glass surface), chemical strengthening is

characterized as per ASTM C-1422 by the magnitudes of compression and the case depth and not by strength.

A comment on measurement of damage during an application should also be made. Although there is considerable understanding of the behavior of chemically strengthened glass to static and *quasi*-static loading conditions, there is insufficient understanding of how strengthened glass would behave under conditions of extreme mechanical shock. Upon being hit by a projectile, the damage is compounded by the tensile stress wave that is reflected from the back surface of the glass. *High-energy impact setups equipped with a high-speed camera and polarized light analysis will be needed for such studies.*

The Technology

One generally uses electrically heated and temperature-controlled stainless-steel tanks. Tanks must be designed properly. Specimens can be suspended using steel cages and proper hoist or conveyor mechanism to lower the load into the tank. Choice of the steel for tank wall and specimen holding cages is important. Oxidizing salts at high temperatures can react with steel causing undesirable contamination from Fe, Ni, and Cr. Care needs to be taken in minimizing steel contact with the glass. Occasionally, Fe from the steel can diffuse into the glass and produce a stain. Thick-walled glass products need to be preheated, and occasionally postheated, to minimize fracture risk from thermal shock.

Continuing use of the salt bath gradually loads the salt with the alkali and Ca ions out of glass. Contamination of the bath with glass ions weakens the potency of the bath due to the rapidly decreasing thermodynamic activity of the invading ions on the glass surface.³⁹ Likewise, adsorption of Ca^{2+} on the glass surface is understood to have a blocking effect to ion exchange. When the bath becomes loaded to an unacceptably high level of contaminants, the molten salt needs to be bailed out and replaced with fresh salt. Some unconfirmed results suggest^{6,40,41} the beneficial use of KOH, etc., added to a KNO_3 bath as scavengers, for instance. Industry practice also indicates the deleterious effects of a humid environment over the bath. Bubbling of SO_2 through the bath to neutralize the pH from $-\text{OH}$ contamination may be useful. Exact mechanisms of strength degradation due to contaminants or improvements due to minor additions to the salt are not understood. *More*

than likely, such additions tend to form complexes with the unknown contaminants in technical-grade salts from local sources.

Attention should be paid to tank operator safety. Proper head-to-toe-covering heat-resistant clothing and safety helmets fitted with eye-protecting visor are a must. Additionally, there are some environmental concerns associated with a chemical strengthening operation. Alkali nitrates are strong oxidizers. Effluent water used for the washing of glass parts should be neutralized before discharging into the sewer stream. Nitrate solid waste requires careful disposal in accordance with local laws.

Using finite-element analysis, it was predicted by Sane and Cooper²⁷ that a 90° corner of a glass plate will become dog-ear shaped following chemical strengthening. Recently, the author's group⁴² has reproduced these results (see Fig. 8) and confirmed the same using non-contact profilometry. Additionally, it was brought to attention that the in-plane stresses are substantially lower in the corner region. This implies that *the edge is much weaker relative to the body of the glass*. Edge should be rounded in plate specimens before the strengthening process.

Another aspect of concern occasionally is the dimensional swelling of the product. In those glass products where the wall thickness is 100 or more times the invading ion penetration distance, the enforcement of

compatibility criteria would leave the glass dimensions showing little swelling. However, in thin products such as the thin display windows in personal communication devices, such dimensional changes could be significant; a provision for such swelling must be made in the manufacturing process (of course, a glass product that had been rapidly cooled and subsequently undergoing prolonged ion exchange treatment near the glass transition could also experience dimensional contraction).

Although multistep strengthening technologies such as thermal tempering plus chemical strengthening using salt-mixed pastes and salt sprays have been tried (for a summary, see Bartholomew and Garfinkel⁶), these technologies are generally of limited use. A novel technique to improve the strength predictability (increasing Weibull modulus) is to develop an “engineered stress profile”⁴³ by deliberately producing a subsurface compression maximum, which maintains a certain minimum stress gradient. The subcritical crack-growth velocity of flaws that are produced on the surface meets increasing resistance, and hence grow only up to the location of the compression maximum. Failure occurs spontaneously when the applied stress intensity factor exceeds the value corresponding to this depth, that is, the glass fails in a narrow distribution of applied stresses (high Weibull modulus). The subsurface compression

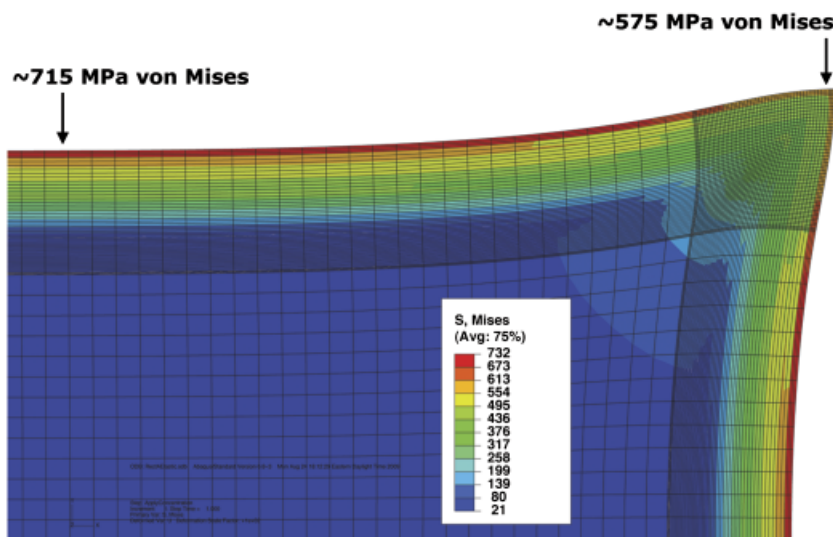


Fig. 8. Two-dimensional plane-strain finite element elastic analysis of the development of stress (MPa) and deformation after $\text{Na}^+ \leftrightarrow \text{K}^+$ ion exchange strengthening in a cell-phone display window glass. Glass plate was 50mm wide $\times$ 0.5 mm (semi-infinite in the third dimension) with 90° corners. Assumed K^+ penetration = 30 μm and a network dilatation coefficient to match the observed surface compression. Only a symmetric quarter view is shown. Note the development of a “dog-ear” and the lowering of surface compression at the corner relative to that in the middle. After Kreski and Varshneya.⁴²

maximum is introduced efficiently by a back exchange in a mixed alkali-salt bath at a lower temperature for a short time. As it occurs, this also means that surface compression is reduced somewhat. The technique may have some use due to its ability to “specify” glass strength, particularly, for those glasses where the case depth is developed to $\sim 50\ \mu\text{m}$ or more.

For effective strengthening by chemical means, glass surface should be as pristine as possible. If flaws are accessible to penetration by salt, the surface would still be strengthened relative to what the strength of the flawed surface would have been.⁴⁴

An interesting application of the chemical strengthening technology is to develop frangible glass products. As stated above, the center tension in an ion-exchanged product of traditional wall thicknesses is of low magnitudes. It is, however, possible to use aluminosilicate glasses where the ion exchange has been carried out to high surface compression and large case depth levels giving rise to a high internal tension. This gives the opportunity to cause franging of the glass after penetrating through the compression case using a sharp needle. This is much like dicing thermally tempered glass; except that whereas diced pieces of a thermally tempered glass are usually 1/2 in. in size, chemically strengthened glass can be franged to submillimeter size. The particle size is clearly a function of the stored elastic energy that can be released from puncturing.^{45–47} The frangible glass technology has some uses in the military.

Conclusions

The author foresees the application of the glass chemical strengthening technology to transparent armor, hurricane-resistant architectural windows, solar energy collector substrates, and needle-free drug delivery cartridges as “just around the corner.” For the armor, in particular, the kinetics of glass damage under extreme loading conditions need to be studied.

To move the technology to expanded applications involving large-scale use of lightweight, high strength-to-weight ratio glass containers, and architectural windows, three specific topics of research need to be pursued. These topics relate to the kinetics of alkali interdiffusion, stress development around an atomic site following ion exchange, and the overall relationship of how surface compression translates to meaningful strengthening, hardening, toughening, and low brittleness.

We have learnt much about the kinetics of the ion exchange. What needs to be learned in an expanded way is a *systematic* study of the effect of minor modifications to the common soda-lime glass compositions upon the alkali interdiffusion coefficient.

Only the rudimentary elements of the stress generation and relaxation science are understood. Cooper’s development of the stress production mathematics by analogy to thermal stress works only in principle and not in the magnitude. The most likely cause of this failure is the onset of network plasticity around each alkali ion exchange site with the pair having a substantial size disparity. We need to know how the atomic polyhedra connected with corner constraints and degrees of freedom affect (i) the hydrostatic yield strength, (ii) the shear yield strength, and (iii) the susceptibility to fracture. With regard to the yield strength estimated on a mesoscopic scale, its translation to the atomic site is yet to be learned. Once the beneficial stress is developed, we need to know what nonviscosity mechanisms cause its relaxation at application temperatures well below the glass transition range.

Acknowledgments

My involvement with chemical strengthening of glass goes back to my employment with Ford Scientific Laboratories, Dearborn, MI, in the early seventies and becoming the usual (objective) reviewer of papers on this topic by my former professor, the late Alfred R. Cooper, Jr. I learnt much during hours of critical discussions of these papers. Research by several of my own students at Alfred University, both undergraduate and graduate, has also helped in developing many of the concepts discussed above. The more notable among these are Vijay Jain, Vineet Tyagi, and Patrick Kreski. Thanks are also due to my colleague Prof. Bill Lacourse for several years of collaboration on the technology of ion exchange.

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